

Table 1: Classification errors (%) on datasets with missing values

Data set	θ	SVM	LRMC	LRMC+SVM	SRMC	VMC-2	VMC-3	NLMC	PMC-S	PMC-W
Mice protein	10%	8.96	6.33	0.8	2.96	0.54	0.46	0.44	0.41	0.39
	50%	32.5	18.78	5.26	13.19	1.24	0.87	0.81	0.71	0.63
Shuttle	10%	12.18	24.7	2.48	21.06	9.7	7.72	4.8	2.66	3.86
	50%	17.82	28.7	10.6	27.3	13.4	11.1	9.58	8.02	9.16
Dermatology	10%	4.48	4.54	3.28	4.21	3.17	3.12	3.08	2.84	2.84
	50%	13.07	9.95	8.83	9.08	8.64	8.74	8.31	8.16	7.98
Satimage	10%	39.6	23.34	14.38	20.62	17.32	15.5	14.7	13.06	14.24
	50%	44.2	24.24	16.96	24.1	18.44	16.18	15.84	14.82	15.18

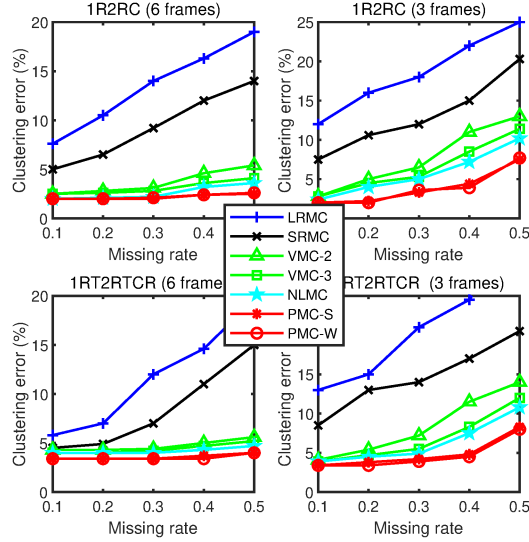


Figure 5: Clustering errors of SSC on Hopkins 155 data recovered by different matrix completion methods

$RAE := \sum_{(i,j) \in \Omega} |X_{ij} - \hat{X}_{ij}| / \sum_{(i,j) \in \Omega} |X_{ij}|$. The average results of 10 repeated trials are reported in Figure 6. The recovery errors of our PMC-S and PMC-W are much lower than those of other methods.

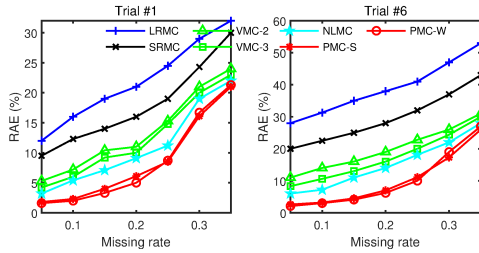


Figure 6: RAE on CMU motion capture data

7.4 Transductive learning on incomplete data

We consider four real datasets with missing values. More details and results are in the supplement. For each dataset, we randomly remove a fraction (denoted by θ) of the observed entries of the feature matrix and perform classification. The

proportion of training (labeled) data is 50%. We use matrix completion to recover the missing entries and classify the data simultaneously. The classification errors (average of 20 repeated trials) are reported in Table 1, in which the least classification error in each case is highlighted by bold. The classification error of SVM with zero-filled missing entries is very high if the missing rate is high. With the pre-processing of LRMC, the classification error of SVM can be significantly reduced. The classification errors of LRMC and SRMC are quite high because they are linear methods that are not effective in nonlinear classification. Our PMC-S and PMC-W outperform other methods in almost all cases.

8 Conclusion

In this paper we studied the problem of high-rank matrix completion and proposed a new model of matrix completion, which is based on analytic functions. We proposed two matrix completion methods PMC-S and PMC-W that outperformed state-of-the-art methods in subspace clustering with missing data, motion capture data recovery, and incomplete data classification. We analyzed the mechanism of the non-linear recovery model and verified the superiority of rank minimization in kernel-induced feature space over rank minimization in data space theoretically and empirically. Future work may focus on improving the scalability of PMC.

9 Acknowledgments

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